Reg No.:_______________ Name:__________________________

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester MCA Degree Regular and Supplementary Examination December 2021

Course Code: RLMCA305

Course Name: CRYPTOGRAPHY AND CYBER SECURITY

Max. Marks: 60 Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks. Marks

1	Identify and define with example two classical encryption techniques?	(3)
2	Define Group and Ring with examples	(3)
3	What is Data Encryption Standard?	(3)
4	How authenticity and confidentiality achieved in encryption by using public key?	(3)
5	What are the properties of hash functions?	(3)
6	What is bitcoin script? Explain its processing procedure.	(3)
7	What is Encapsulating Security Payload (ESP)?	(3)
8	What is Non-Repudiation? Explain Non-Repudiation based on Public Key Technology.	(3)

PART B

Answer six questions, one full question from each module and carries 6 marks.

Module I

9	a)	Explain active attacks and passive attacks.	(3)
	b)	Given the key 'MONARCHY', apply the Playfair cipher to the plaintext 'FACTIONAISM'. Decrypt the ciphertext also.	(3)

OR

10	a)	Explain one-time pad. What are the two problems with the one-time pad?	(3)
	b)	Explain the symmetric cipher model.	(3)

Module II

11	State and prove Euler's theorem.	(6)

OR

12	Explain the Miller- Rabin algorithm for testing primality.	(6)

Module III

13 With a suitable diagram explain Cipher Block Chaining (CBC) mode. (6)

OR

14 Discuss the different attacks on RSA. (6)

Module IV

15 What do you mean by Message Authentication? Explain Cipher-Based Message Authentication code (CMAC). (6)

OR

16 Explain Digital Signature Algorithm (DSA) in detail. (6)

Module V

17 a) Explain Scrooge Coin. (2)

b) What are the two kinds of transactions in Scrooge Coin? (4)

OR

18 Explain bitcoin transactions. (6)

Module VI

19 a) Explain S/MIME. (3)

b) Explain Exportability in SSLv2 and Exportability in SSLv3. (3)

OR

20 Explain Object formats and Primitive Object formats in PGP. (6)

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Reg No.:_______________ Name:__________________________

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester MCA Regular and Supplementary Examination December 2020

Course Code: RLMCA305

Course Name: CRYPTOGRAPHY AND CYBER SECURITY

Max. Marks: 60 Duration: 3 Hours

PART A

Answer all questions, each question carries 3 marks. Marks

1	Illustrate symmetric cipher model in cryptography.	(3)
2	Compute the multiplicative inverse of 23 in Z_{100}.	(3)
3	Write short note on modern stream ciphers.	(3)
4	Compare and contrast symmetric and asymmetric key encryption.	(3)
5	List out the classification of digital signature schemes.	(3)
6	Discuss the features of Bitcoin.	(3)
7	Illustrate PGP message format in Email security.	(3)
8	Explain security services for Email.	(3)

PART B

Each question carries 6 marks.

Module I

9	Explain security services and mechanisms in cryptography.	(6)
	OR	
10	Explain steganography method with text covering process.	(6)

Module II

11	Explain Euclidean algorithm and find the multiplicative inverse of 11 in Z_{26} using the algorithm.	(6)
	OR	
12	Give a short note on the following. i) Fermat's theorem ii) Euler's theorem iii) Testing for Primality	(6)

Module III

13 Define Asymmetric key Cryptography and explain about the RSA cryptosystem with suitable example. (6)

OR

14 Explain Data Encryption Standard with a neat diagram. (6)

Module IV

15 Define hash functions? What is its significance? (6)

OR

16 Describe various attacks on digital signature. (6)

Module V

17 Discuss the process of storing and usage of Bitcoins in network. (6)

OR

18 Illustrate and explain Bitcoin transaction in network. (6)

Module VI

19 Discuss the protocols of SSL. (6)

OR

20 Explain two modes of IPSEC. (6)

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Reg No.:_______________ Name:__________________________

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester MCA Degree (S, FE) Examination May 2024

Course Code: RLMCA305

Course Name: CRYPTOGRAPHY AND CYBER SECURITY

Max. Marks: 60 Duration: 3 Hours

PART A

Answer all questions, each carries3 marks. Marks

1	Discuss Non- Repudiation in detail	(3)
2	State and Explain Euler's Theorem	(3)
3	Discss the CBC mode of block cipher with neat diagram	(3)
4	Write note on Elliptic-curve Cryptography	(3)
5	Illustrate Blind Signature scheme in detail	(3)
6	Describe the working of the Scrooge Coin	(3)
7	Differentiate transport and tunnel mode in IPSec	(3)
8	List out major Email security challenges	(3)

PART B

Answer six questions, one full question from each module and carries 6 marks.

Module I

9	Discuss any two poly-alphabetic substitution techniques with suitable examples	(6)
	OR	
10	Explain different various security services in detail	(6)

Module II

11	State and Explain Chinese Reminder Theorem with suitable example	(6)
	OR	
12	Explain finite fields of the form $GF(2^n)$ with examples	(6)

Module III

13	Explain AES algorithm in with a neat diagram.	(6)
	OR	
14	Explain Diffie-Hellman key exchange algorithm with suitable diagrams	(6)

Module IV

15	Explain RSA Digital Signature scheme and attacks on RSA Digital Signature schemes	(6)

OR

16	Define the concept of Message Authentication and Explain Cipher-Based Message Authentication code (CMAC) in detail	(6)

Module V

17	Explain different storage mechanisms for Bitcoin	(6)

OR

18	Explain how Splitting and Sharing of Keys achieved in Bitcoin network	(6)

Module VI

19	Describe the working of secure electronic transaction with suitable diagrams	(6)

OR

20	Outline the IPSec Encapsulating Security Payload in detail.	(6)

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